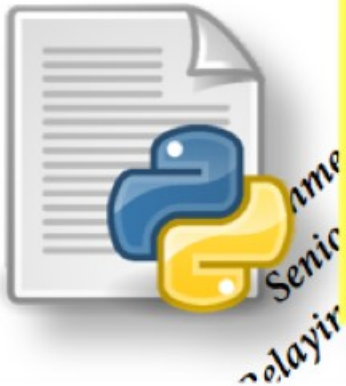
Python Installation and Setup.



Python Installation

Virtual Environment Setup / Creation

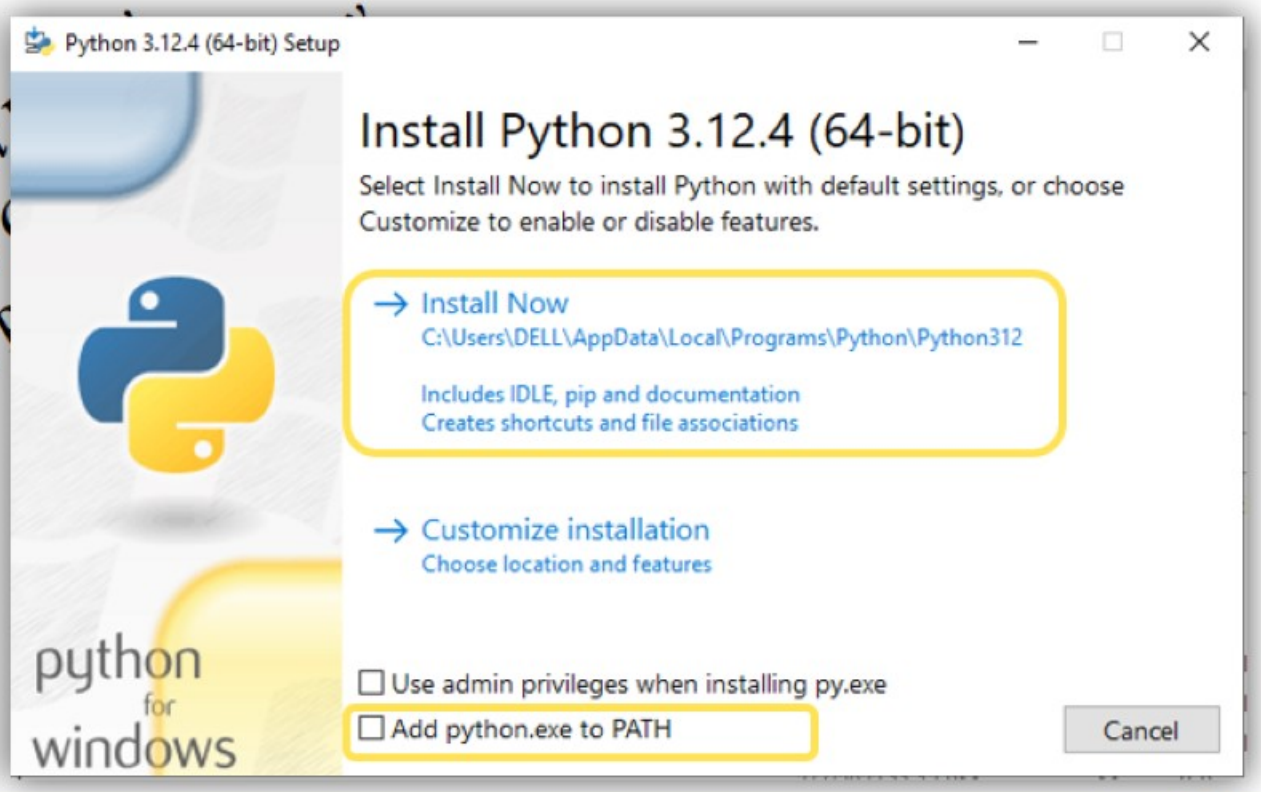
Integrated Development Environment (IDE)

Ahmed Khairy
Senior Geophysicist
Belayim Petroleum Company

Python Installation

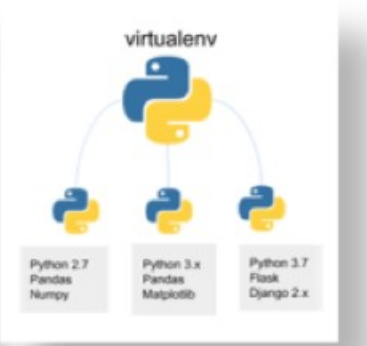
- ❑ **Install Python for Windows.**
- ❑ **Visit Python Website.**

  https://www.python.org/downloads/windows/
- ❑ **Download and Install windows installer.**

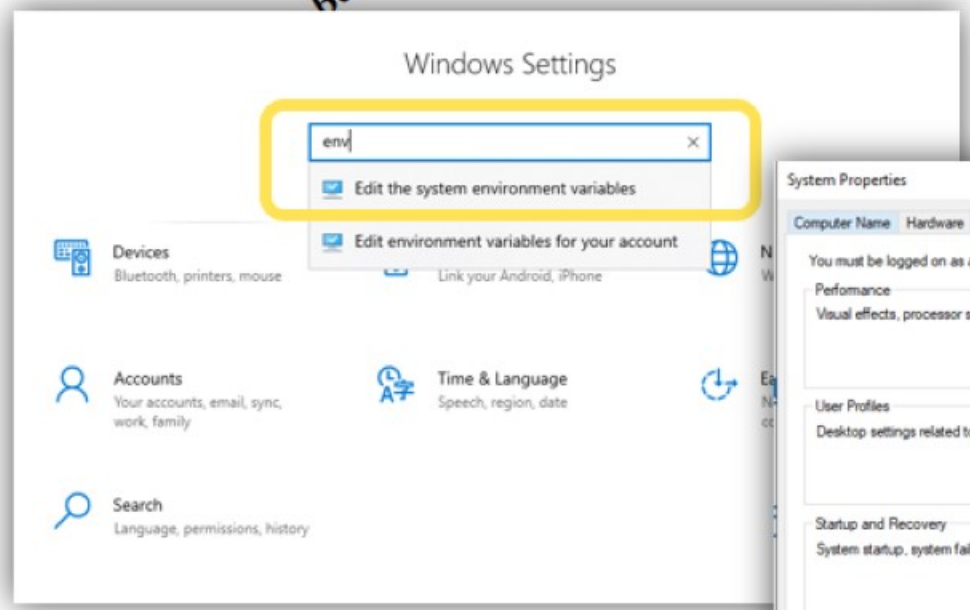


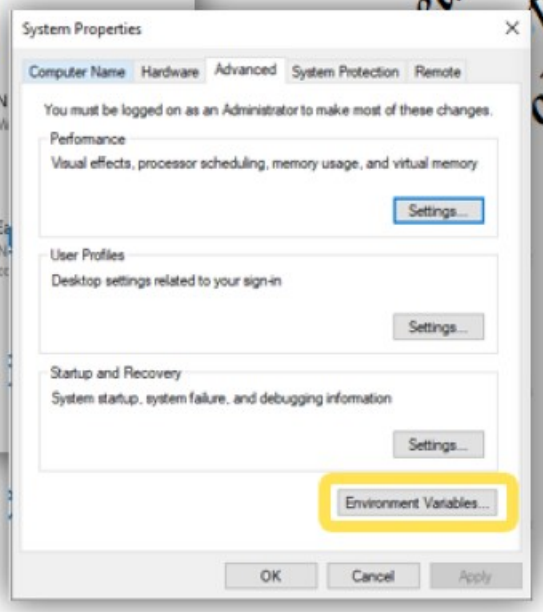
Python Installation and Setup.

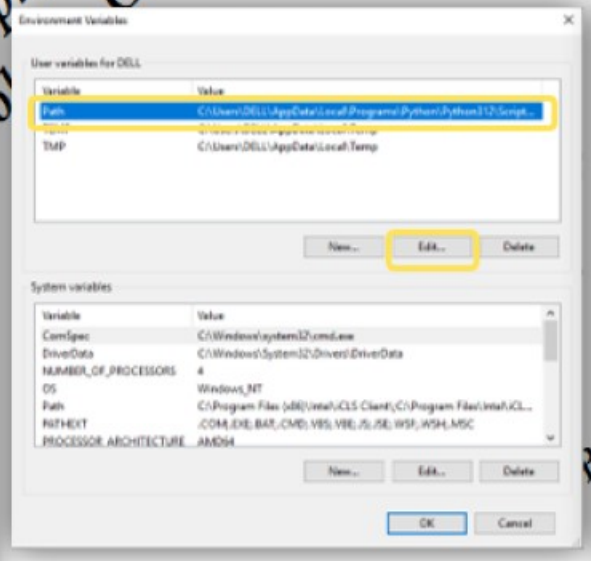
Virtual Environment Setup / Creation

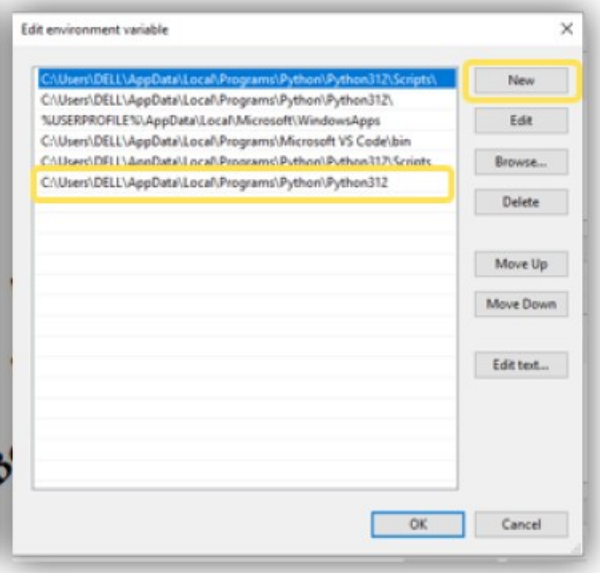


- In case of not adding python.exe to the environment path during installation, setup should be done manually



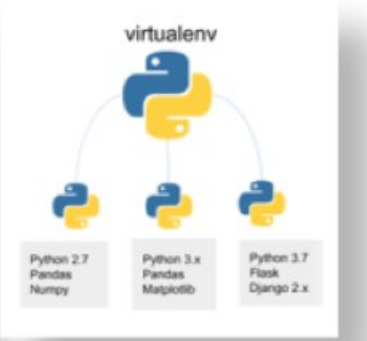




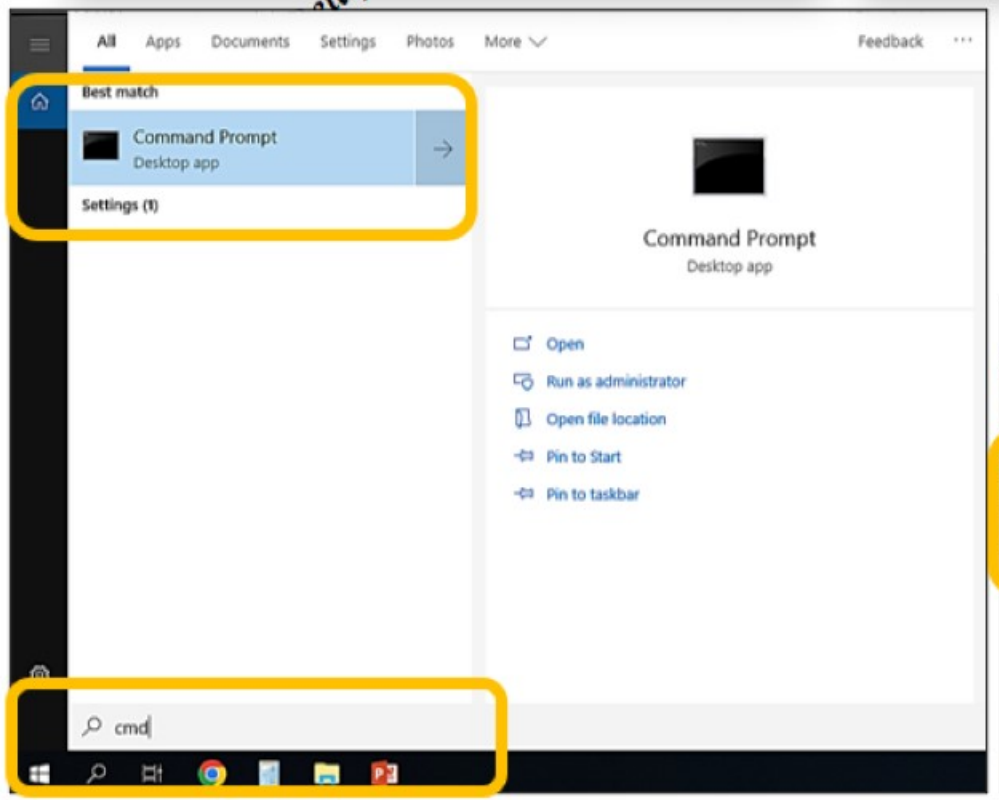


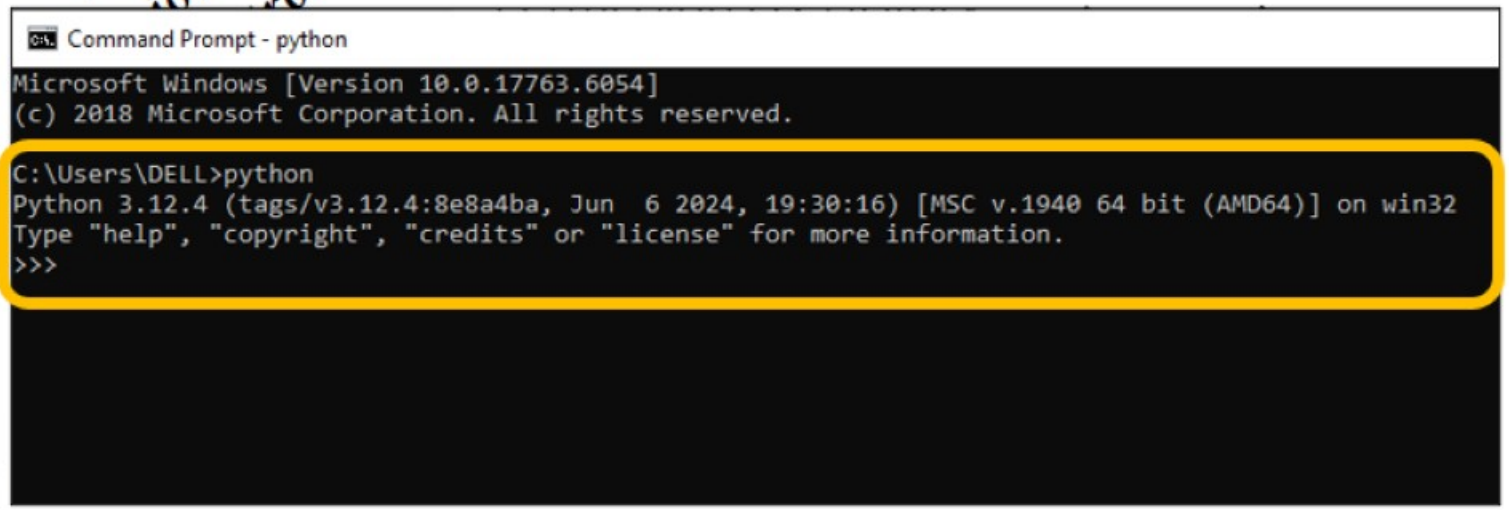
Python Installation and Setup.

Virtual Environment Setup / Creation



- In case of not adding python.exe to the environment path during installation, setup should be done manually
- Open " cmd " and type " python " to check if the interpreter environment successfully identified.





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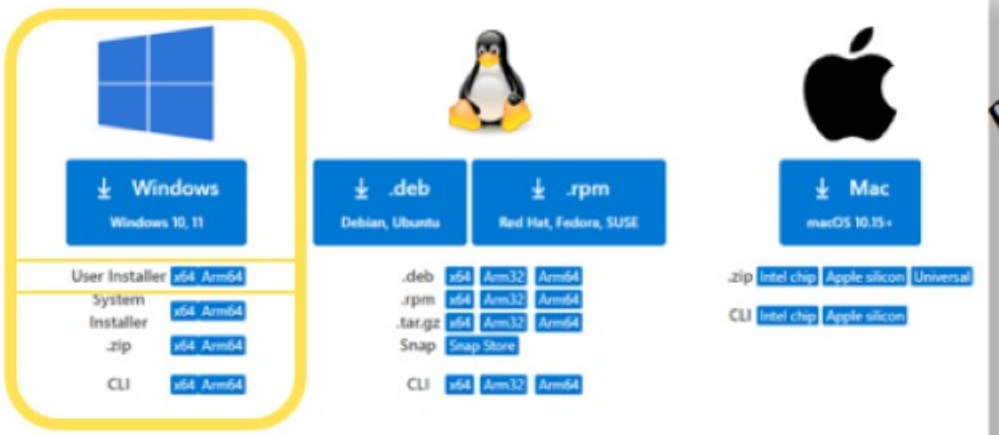
Petroleum Company

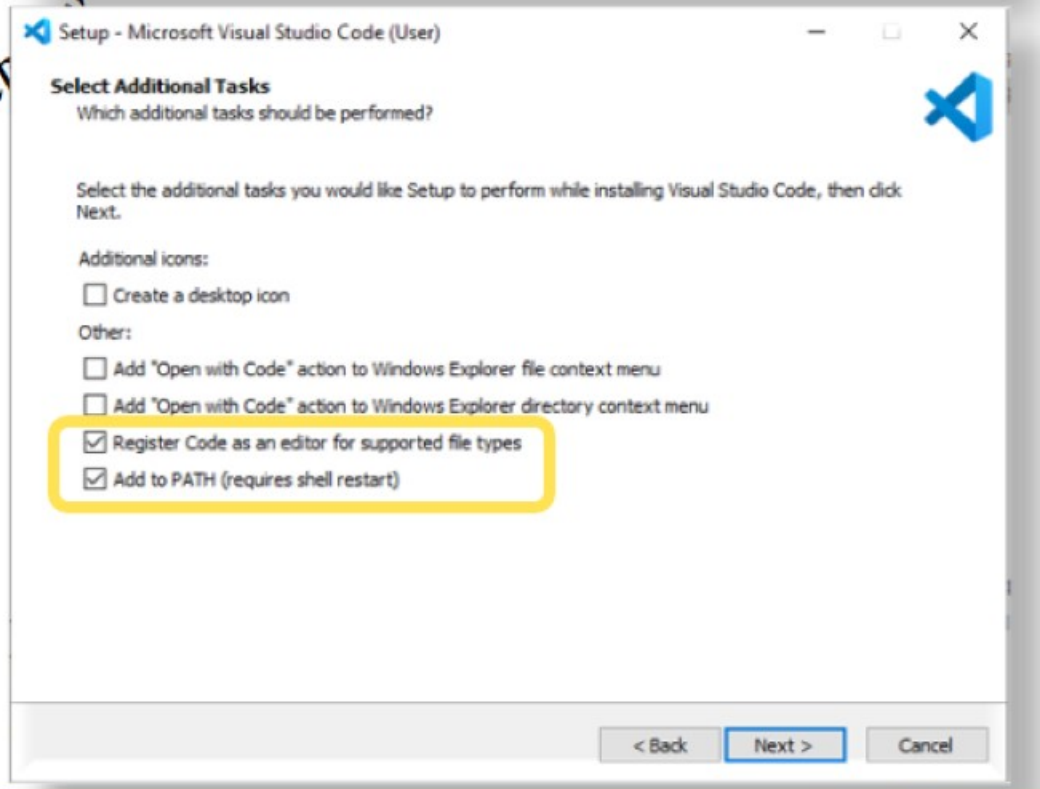
Python Installation and Setup.

Integrated Development Environment (IDE)

- ❑ **Install VS Code for Windows.**
- ❑ **Visit Visual Studio Code Website.**

https://code.visualstudio.com/Download





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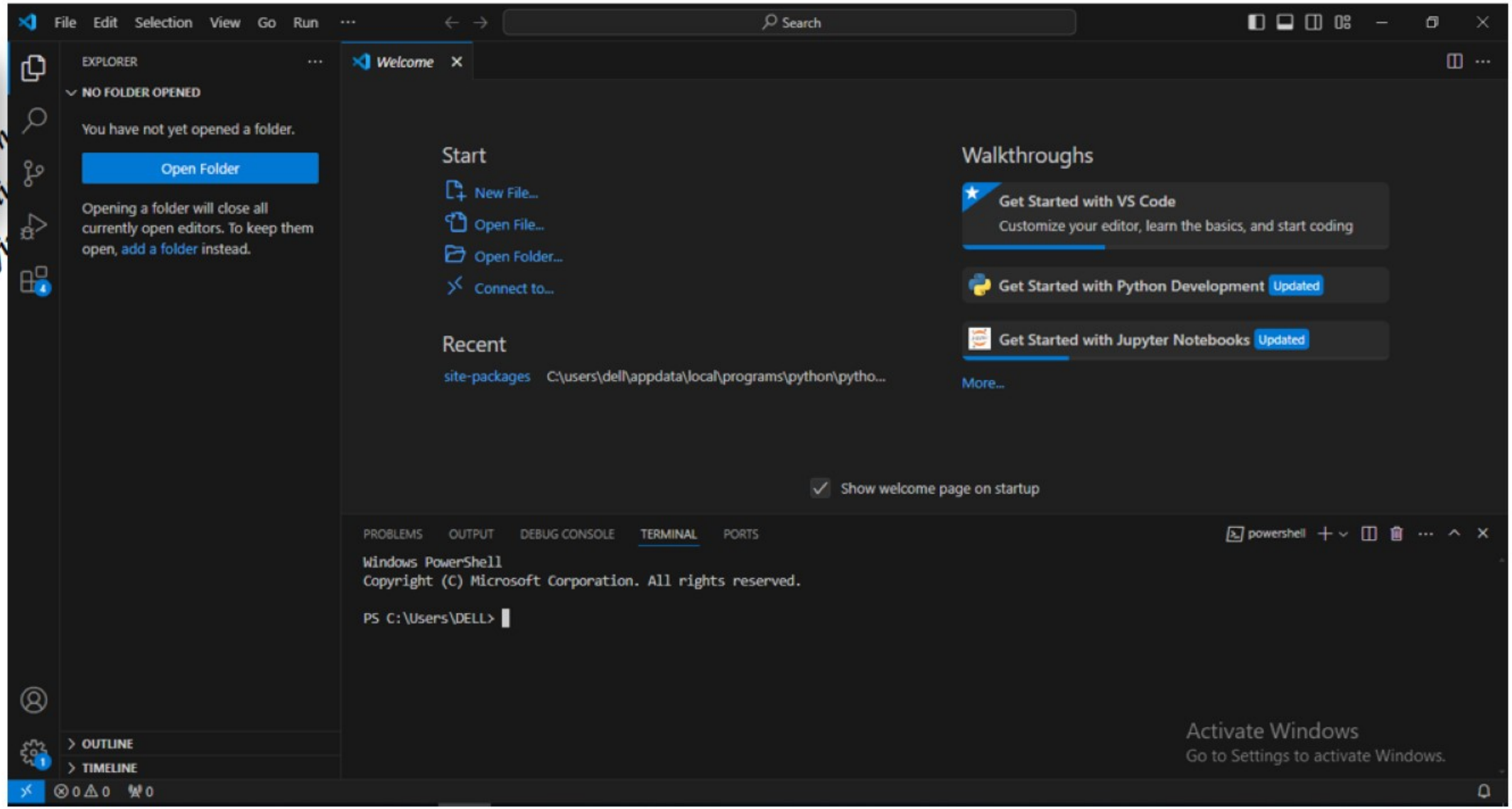
Python Installation and Setup.



Python Installation and Setup.

Virtual Environment Setup / Creation

- Check Python version

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS    JUPYTER

PS C:\Users\DELL> python --version
Python 3.12.4
PS C:\Users\DELL>
```

- List all installed Python version

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS    JUPYTER

PS C:\Users\DELL> py -0p
 -V:3.12 *        C:\Users\DELL\AppData\Local\Programs\Python\Python312\python.exe
PS C:\Users\DELL>
```

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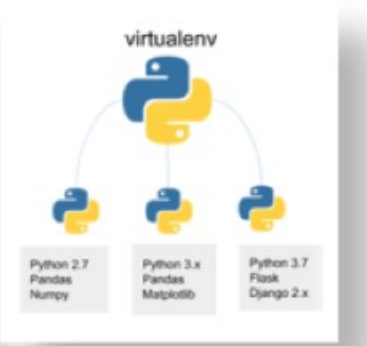
Virtual Environment Setup / Creation



❑ Creating a Virtual Environment

1. Create a folder for the new envs. Documents

```
PROBLEMS   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS   JUPYTER

PS C:\Users\DELL> New-Item -Path 'C:\Users\DELL\Venv_Create' -ItemType Directory

    Directory: C:\Users\DELL

Mode                LastWriteTime         Length Name
----                -------------         ------ ----
d-----        8/9/2024   3:09 PM                Venv_Create
```

2. Access the created environment folder

```
PROBLEMS   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS   JUPYTER

PS C:\Users\DELL> cd C:\Users\DELL\Venv_Create\
PS C:\Users\DELL\Venv_Create>
```

3. Create new env. Called “new_py_env”

```
PROBLEMS   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS   JUPYTER

PS C:\Users\DELL\Venv_Create> python -m venv new_py_env
```

4. List the content of the new env. folder

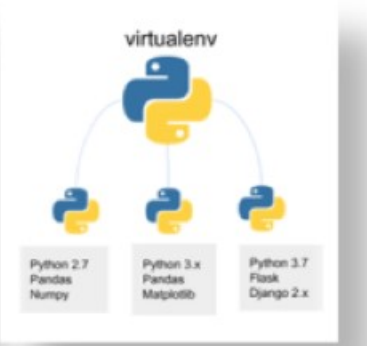
```
PROBLEMS   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS   JUPYTER

PS C:\Users\DELL\Venv_Create> ls

    Directory: C:\Users\DELL\Venv_Create

Mode                LastWriteTime         Length Name
----                -------------         ------ ----
d-----        8/9/2024   3:20 PM                new_py_env

PS C:\Users\DELL\Venv_Create>
```

New folder with the new environment name is created.

Belayim
Ahmed K
Senior G
Belayim P
Ahmed K
Senior Geophys
Belayim Petroleum Co

Python Installation and Setup.

Virtual Environment Setup / Creation



❑ Creating a Virtual Environment

5. List the content of the new env. (new_py_env)

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS    JUPYTER

PS C:\Users\DELL\Venv_Create\new_cmd\Scripts> cd C:\Users\DELL\Venv_Create\new_py_env\
PS C:\Users\DELL\Venv_Create\new_py_env> ls


    Directory: C:\Users\DELL\Venv_Create\new_py_env


Mode                LastWriteTime         Length Name
----                -------------         ------ ----
d-----         8/9/2024   3:20 PM                Include
d-----         8/9/2024   3:20 PM                Lib
d-----         8/9/2024   3:20 PM                Scripts
-a----         8/9/2024   3:23 PM            318 pyvenv.cfg
```

6. Activate the environment (in cmd)

```
Administrator: Command Prompt
Microsoft Windows [Version 10.0.17763.6054]
(c) 2018 Microsoft Corporation. All rights reserved.

C:\Windows\system32>cd C:\Users\DELL\Venv_Create\new_py_env\Scripts

C:\Users\DELL\Venv_Create\new_py_env\Scripts>activate

(new_py_env) C:\Users\DELL\Venv_Create\new_py_env\Scripts>
```

7. Install (jupyterlab) and (notebook)

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS    JUPYTER

PS C:\Users\DELL\Venv_Create\new_py_env> .\pip.exe install jupyterlab
```

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS    JUPYTER

PS C:\Users\DELL\Venv_Create\new_py_env> .\pip.exe install notebook
```

Ready to Code !